

# Monte Carlo Integration

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## Monte Carlo Integration

Monte Carlo Integration is a statistical method based on random sampling

Idea:

- If  $X$  is a random variable with pdf  $f(x)$ , then

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

- Thus, to estimate  $\omega = \int_a^b g(x)dx$ , we do

$$X_1, \dots, X_n \sim U(a, b), \text{ then } \hat{\omega} = \frac{1}{n} \sum_{i=1}^n g(X_i)$$

## Algorithm

- 1) Generate iid  $X_1, \dots, X_n$  from  $U(a, b)$
- 2) Compute  $g(\bar{X}) = \frac{1}{n} \sum_{i=1}^n g(X_i)$
- 3) then  $\hat{\omega} = (b - a)g(\bar{X})$

## Example

Compute a Monte Carlo Estimate of  $\omega = \int_0^1 e^{-x} dx$

```
set.seed(123)
n<-10000

x <- runif(n, min=0, max=1)
f_x <- exp(-x)

#Average of g(X)
omega_hat <- mean(f_x)

#compute (b-a)mean(g(X))
(1-0)*omega_hat
```

```
## [1] 0.6333057
```

Compute a Monte Carlo Estimate of  $\omega = \int_2^4 e^{-x} dx$

```
set.seed(123)
n<-10000

x <- runif(n, min=2, max=4)
f_x <- exp(-x)

#Average of g(X)
omega_hat <- mean(f_x)

#compute (b-a)mean(g(X))
(4-2)*omega_hat
```

```
## [1] 0.117334
```

Estimate  $\phi(-1.645) = \int_{-\infty}^{-1.645} \frac{1}{\sqrt{(2\pi)}} e^{-\frac{t^2}{2}} dt$  using Monte Carlo Method

- Generating  $n$  random samples  $X_1, \dots, X_n$  from a standard normal distribution  $N(0, 1)$
- Counting how many samples satisfy  $X_i \leq -1.645$
- Computing the proportion:

$$\phi(-1.645) = \frac{\# \text{ of } X_i \leq -1.645}{n}$$

```
set.seed(123)
n<-10000

samples <- rnorm(n)

count <- sum(samples <= -1.645)

estimate <- count/n
estimate # or estimate <- mean(samples <= -1.645)
```

```
## [1] 0.0499
```

## Importance Sampling

Suppose we want to estimate:

$$\hat{\mu} = E[g(X)] = \int g(x)f(x)dx,$$

where  $X \sim f(x)$ , and  $f(x)$  is the density of  $X$ . We can rewrite the expectation using another density  $h(x)$  (the importance distribution) as long as  $h(x) > 0$  whenever  $h(x)f(x) \neq 0$  (to avoid division by zero):

$$\mu = \int h(x)f(x)dx = \int g(x)\frac{f(x)}{h(x)}h(x)dx = E_h[g(x)\frac{f(x)}{h(x)}]$$

where the expectation is now taken over  $X \sim h(x)$

## Algorithm

The importance sampling estimator is:

1. Draw  $n$  samples  $X_1, \dots, X_n \sim h(x)$
2. Compute the weights  $\omega(X_i) = \frac{f(X_i)}{h(X_i)}$
3. Estimate  $\mu$  as:

$$\mu = \frac{1}{n} \sum_{i=1}^n g(X_i) \frac{f(X_i)}{h(X_i)}$$

## Example

Simulate  $\phi(-2) = P(Z \leq -2)$  using importance sampling with  $h(x) \sim \text{Cauchy}(1)$

```
set.seed(123)
n<-10000

x <- rcauchy(n)

f <- function(x){
  return(dnorm(x))
}

h <- function(x){
  return(dcauchy(x))
}

g <- function(x){
  return(x <= -2)
}

estimate <- mean(g(x) * (f(x)/h(x)))
estimate
```

```
## [1] 0.02253659
```